TEXORA · LECTURE SHEET

Fibre Migration

Fibres do not stay where spinning puts them: during twisting they travel radially in and out of the yarn body — radial migration — and in blends some components systematically claim the core while others take the surface. The mechanisms, the measurement indices, and the consequences for appearance, dyeing and wear.

TE-3209 · TEXTILE PHYSICS · TOPIC 8 · LEVEL 3 TERM 2

1. What migration is

A spun yarn is not a frozen rope of parallel fibres. A tracer fibre followed along a yarn is found now at the surface, now buried in the core, oscillating radially as it advances — **radial migration**. It is created in the spinning triangle: fibres are gripped, drafted and twisted under varying tension; a fibre under higher tension pulls toward the axis (the shortest path), displacing a slacker neighbour outward, and roles reverse along the length as tensions fluctuate. The result is a helical path of varying radius — and this continual interlocking is a major part of why a staple yarn holds together at all: migrating fibres mechanically stitch the structure, supplementing friction (Topic 5).

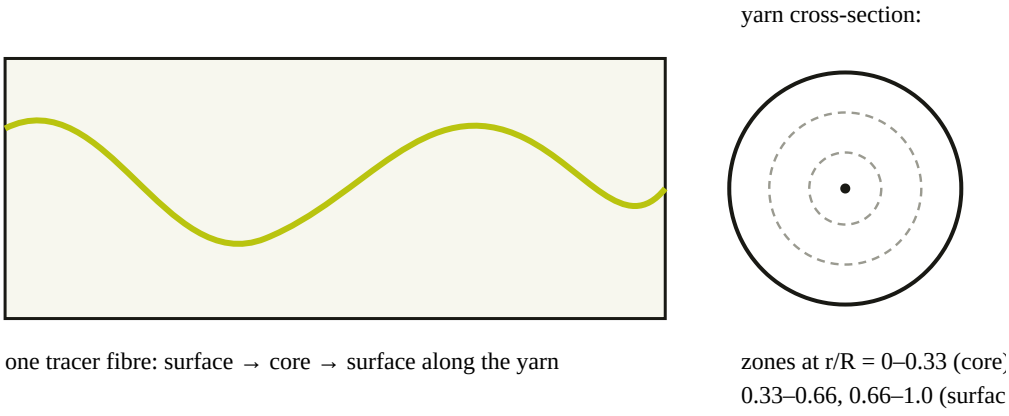


Figure 1 — Left: a tracer fibre’s radial migration along the yarn length. Right: the radial zones of the cross-section across which migration and blend distribution are counted.

2. Preferential migration in blends

In a blend, the components do not migrate equally. Systematic evidence (Morton, Hearle and co-workers) ranks the drivers:

- **Length:** longer fibres migrate *inward* (they are held at both ends and tensioned into the core).
- **Fineness:** finer fibres migrate inward — they bend easily into the tight core paths.
- **Stiffness/initial modulus:** stiffer, higher-modulus fibres resist tight curvature and stay outward.
- **Cross-section and crimp** modify the packing: bulky/crimped fibres occupy the outer, lower-density zones.

So in polyester–cotton, the longer, finer component claims the core; in wool blends the coarser or stiffer partner wraps the outside. The surface composition of a blend yarn therefore **differs from its bulk composition** — and since dyeing, abrasion, pilling, handle and appearance are all surface events, preferential migration is a design variable and a defect source in one.

> **WORKED EXAMPLE — A MEAN FIBRE POSITION INDEX**
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> Count component A's fibres across yarn cross-sections in five radial zones centred at r/R = 0.1, 0.3, 0.5, 0.7, 0.9, obtaining counts 5, 15, 30, 30, 20 (total 100). Mean position = (0.1×5 + 0.3×15 + 0.5×30 + 0.7×30 + 0.9×20)/100 = (0.5 + 4.5 + 15 + 21 + 18)/100 = **0.59**. Against a mid reference of 0.50, A leans to the **surface**; its blend partner, at 0.41 by the same count, occupies the core. Predict the consequences: A dominates the fabric's appearance, dye shade and abrasion exposure out of proportion to its blend share — the standard explanation when a 50/50 blend "behaves like" its minor component at the surface.

3. Measuring and using migration

Hamilton's indices formalise the counting: mean fibre position, radial packing density (fibres per zone area — normally highest mid-radius, looser at the very surface), and migration parameters fitted to tracer paths (amplitude and frequency of the radial oscillation). Tracer techniques (dyed tracer fibres photographed through cleared yarn, or sectioning) remain the reference methods. Engineers spend the results in: **core-spun yarns** (migration driven to its limit — a filament core deliberately 100% buried, elastic or high-tenacity heart with a staple sheath for comfort and dyeing); **blend engineering** (putting the expensive or functional fibre where it works — silver/antistatic at the surface, strength in the core); and **defect diagnosis** — barre-like shade variation, differential dyeing of blended fabrics and premature surface wear are frequently migration distributions made visible.

4. Key points

- Radial migration: fibres oscillate between core and surface along the yarn, driven by tension differences in the spinning triangle; it stitches the yarn together.
- Preferential migration: longer, finer, more flexible fibres go inward; shorter, coarser, stiffer fibres stay out.

- Surface composition ≠ bulk composition — dyeing, appearance, pilling and wear read the surface.
- Mean position index example: 0.59 vs partner 0.41 from simple zone counting (Hamilton's method).
- Core-spun yarn is migration by design; shade and wear defects in blends are often migration by accident.

5. Practice

Answer in your own words; brief model answers follow.

1. In a polyester (long, fine staple) / cotton (shorter, coarser) blend, predict each component's preferred zone and the resulting first-wear behaviour of the fabric surface.
2. Zone counts for component B are 20, 30, 25, 15, 10 at r/R = 0.1…0.9. Find B's mean position and compare with the sheet's component A (0.59).
3. Why does radial migration strengthen a staple yarn rather than merely rearrange it?
4. A blended fabric dyes unevenly although fibre-level dye affinity is uniform. Offer a migration-based explanation and a test (surface vs bulk composition) that confirms it.
5. Design, in principle, a yarn for a shirt that should look and feel like cotton but dry and endure like polyester, using migration deliberately.
6. Why might raising spinning tension change the blend's surface composition?

Answer key

1. Polyester (longer, finer) migrates inward; cotton takes the surface. The shirt therefore looks, feels and dyes like cotton at first — but as cotton surface fibres abrade away, the polyester core shows through progressively: shade lightens and a synthetic sheen appears with wear (the classic blend ageing pattern, here a feature planned for, if the designer counted zones).
2. Mean = (0.1×20 + 0.3×30 + 0.5×25 + 0.7×15 + 0.9×10)/100 = (2 + 9 + 12.5 + 10.5 + 9)/100 = 0.43 — B is core-leaning, consistent with being A's complement (0.59 + 0.43 ≈ 1.02 ≈ symmetry about 0.5, small counting noise aside).
3. A fibre alternating between core and surface crosses and locks with neighbours along its whole length; pull-out then requires defeating friction at many separated contact points and bending the fibre through the structure — migration converts a bundle of parallel fibres into an interlaced solid, so yarn strength exceeds the parallel-bundle prediction especially at low twist.
4. Preferential migration has made the surface richer in the component that takes the dye more strongly (or presents more area), so colour yield varies with local migration even when bulk blend ratio is exact. Test: compare surface fibre counts (microscopy of the yarn exterior) against bulk analysis (dissolution) — a systematic difference convicts migration.

5. Core-spun or core-biased blend: polyester filament/fine long-staple core (strength, moisture transport out, dimensional stability) wrapped in a cotton sheath (surface look, touch, dyeing). Migration rules put the longer/finer synthetic inward naturally; a core-spun construction guarantees it absolutely.
6. Tension is the migration driver: higher spinning tension increases the inward pull on tensioned (longer/finer) fibres and tightens radial packing, enhancing preferential separation — settings are composition tools, not just break-rate knobs.
